1 Android Overview

Android is a software stack for portable devices such as mobiles and tablets. The Android operating system is based on a modified Linux kernel. The Android developer community extends the functionality of the devices beyond the stock applications, and there already are over a hundred thousand Android apps in the Android Market (Google's online Android App store). Coding for these applications is mainly done in Java using Google-developed Java libraries. The Android platform is coded in C for its core, C++ for third-party libraries and Java for the user interface. Most of the code is under the Apache license, as free and open source software. The latest stable release of Android is the Honeycomb (Android 3.0.1) for tablets and Gingerbread (Android 2.3.3) for mobiles. It supports ARM, MIPS, Power and x86 platforms.

Android's open source stack runs on a Java-based, object-oriented application framework. It operates over Java core libraries running on the Dalvik virtual machine (VM). Prior to execution, Android applications are converted into Dalvik executables (DEX) format, rendering them suitable for portable devices with memory and processing speed constraints; as it is a register based architecture, unlike the stack machines of Java VMs. The database management system used is SQLite. Android uses Open Graphics Library for Embedded Systems (OpenGL ES) 2.0 3D graphics application programming interface (API).

1.1 Features

The platform is adaptable to large VGA displays, 2D graphics library and 3D library based on OpenGL. It supports all ranges of connectivity technologies such as: GSM/EDGE, CDMA, EV-DO, UMTS, Bluetooth, Wi-FI and WiMAX. The stock web browser is based on the open source WebKit layout engine coupled with Chrome's V8 JavaScript engine to render web pages. Android supports several media formats such as H.264, MPEG-4, AMR, AAC, MP3, WAV, Ogg, JPEG, PNG and GIF. It supports camera (video and still), touch screen, GPS, accelerometers, gyroscopes, magnetometers, proximity and pressure sensors, thermometers and accelerated 3D graphics; giving an application developer several options to mull over while developing compelling apps.

The software development kit (SDK) comprises an emulator (the Android Virtual Device), debugging tools, memory and performance profiling.